

Objective: The objective of this experiment is to implement flip flops.

Problem-1:

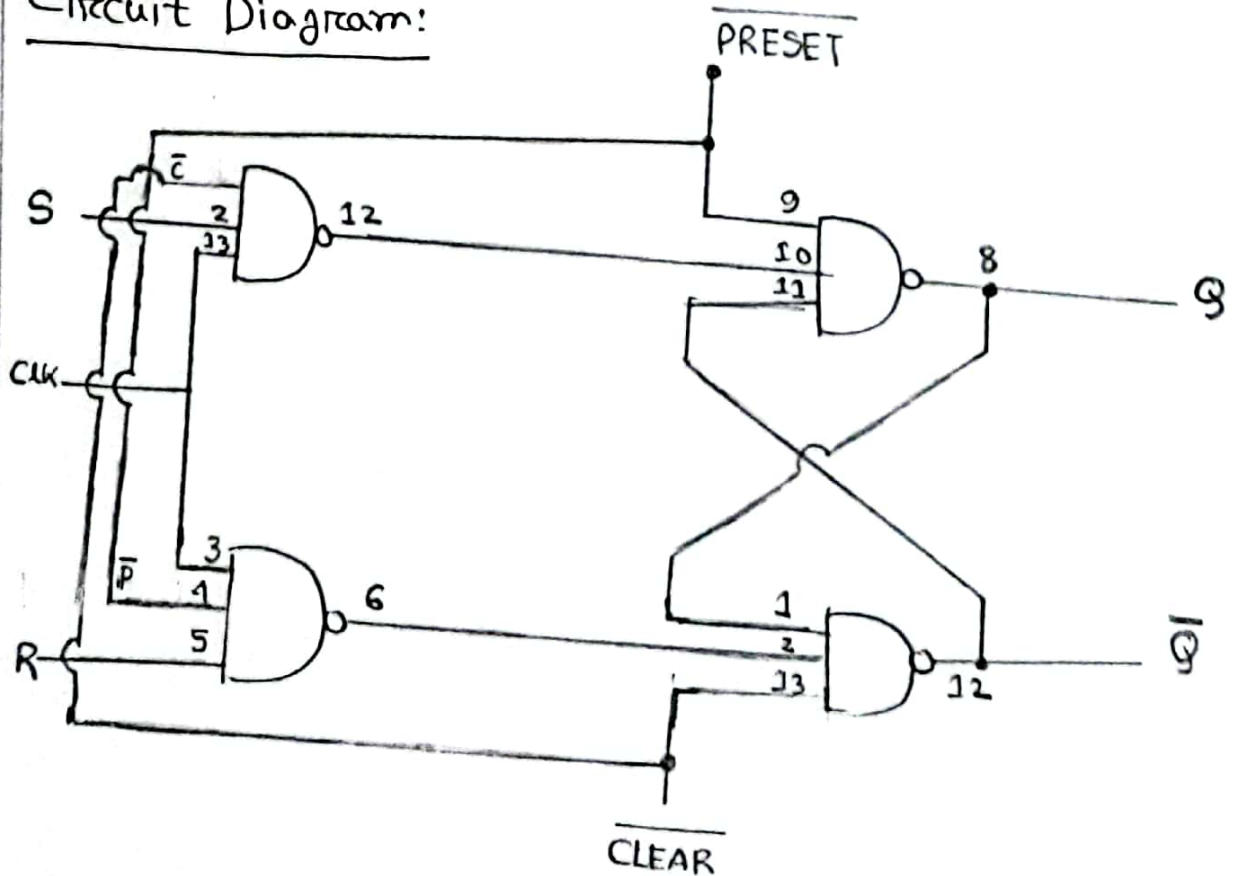
problem specification: We need to design and implement an SR flip flop with $\overline{\text{preset}}$ and $\overline{\text{clear}}$ input, using only nand gates

Required instruments:

<u>item</u>	<u>Quantity</u>
1. Trainer board - - - - -	1
2. IC extractor - - - - -	1
3. 3 Input NAND Gate - - - - - (IC 7410)	2
4. Wires - - - - -	a lot
5. Multimeter - - - - -	1

Excitation table:

$\overline{\text{PRESET}}$	$\overline{\text{CLEAR}}$	CLK	S	R	Q_{n+1}	mode
0	0	X	X	X	Invalid	Not Allowed
0	1	X	X	X	1	set
1	0	X	X	X	0	Reset
1	1	0	X	X	Q_n	No change
1	1	1	0	0	Q_n	No change
1	1	1	0	1	0	Reset
1	1	1	1	0	1	set
1	1	1	1	1	X	Not allowed

Circuit Diagram:Fig: SR flipflop with $\overline{\text{PRESET}}$ & $\overline{\text{CLEAR}}$

Problem-2: We need to design and implement a master slave JK flipflop using only nand gates.

Required instruments:

<u>item</u>	<u>Quantity:</u>
1. Trainer board - - - - -	1
2. IC extractor - - - - -	1
3. 3 input NAND GATE - - - - - (IC 7410)	1
4. 2 input NAND gate - - - - - (IC 7403)	2
5. Wires - - - - -	a lot
6. Multimeter - - - - -	1

Excitation table:

clk	j	k	Q_n	Q_{n+1}	Mode
0	x	x	0	0	Hold
0	x	x	1	1	
1	0	0	0	0	No change
1	0	0	1	1	
1	0	1	0	0	Reset
1	0	1	1	0	
1	1	0	0	1	set
1	1	0	1	1	
1	1	1	0	1	Toggle
1	1	1	1	0	

Circuit diagram:

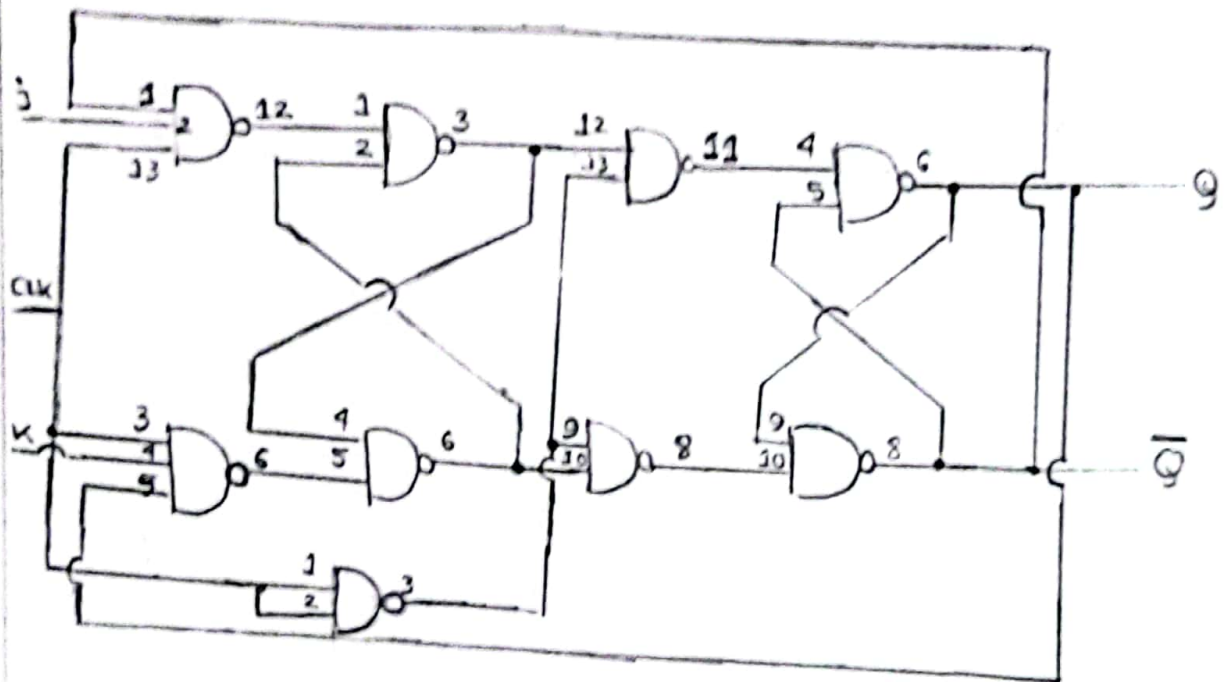


Fig: Master slave JK flipflop